

Bo-Yu Chen 陳柏昱

Researcher and builder working across music AI, creative tools, and human-computer interaction.

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RESEARCH INTERESTS

Human-Computer Interaction (HCI) · Music Information Retrieval (MIR) · AI Agents · Creative Support Tools · Musical Interfaces

EDUCATION

National Taiwan University (NTU) Sep 2021 – Aug 2025
M.S. in Networking and Multimedia (Dept. of CSIE) Taipei, Taiwan

- Master's Thesis: *FlueBricks: A Modular Toolkit for Interactive Flute-like Instrument Systems* [C1] (Advisor: Prof. Lung-Pan Cheng)
- Thesis Honors: Winner of TAICHI 2026 Master's Thesis Honorable Mention Award Top 16.6%

National Taiwan University of Science and Technology (NTUST) Sep 2013 – Jun 2018
B.S. in Electrical Engineering Taipei, Taiwan

HONORS & AWARDS

- Master's Thesis Honorable Mention Award · TAICHI Thesis Award for *FlueBricks* (Top 16.6%, published as [C1]) 2026
- Best Paper Award 🏆 · Taiwan Computer-Human Interaction Society (TAICHI) for *FlueBricks* [C1] 2025
- Overseas Visit Delegate · Epoch Foundation Young Entrepreneurs of the Future (Top 9%, 16/175) 2017
- 3rd Prize · Ministry of Economic Affairs (MOEA) GetFresh Creativity Competition 2017

RESEARCH EXPERIENCE

Computational Physicality Lab, National Taiwan University Sep 2021 – Aug 2025
Graduate Research Assistant (Advisor: Prof. Lung-Pan Cheng) Taipei, Taiwan

- Designed **FlueBricks**, a modular physical-computing toolkit enabling acoustic replication and customizable design of flute-like instruments as Master's thesis research. [C1]
- Engineered parametric CAD assemblies in **Fusion 360** for digital fabrication, and conducted user evaluations across musician skill levels to validate expressive playability.

Creative AI Lab, Sony Group Corporation Nov 2020 – Sep 2021
Research Intern (Advisors: Dr. Yuki Mitsufuji, Dr. Marco A. Martínez Ramírez, Wei-Hsiang Liao) Tokyo, Japan (Remote)

- Developed **DJTransGAN**, the first generative model for automated cross-track DJ transitions using DDSP and GANs. [C2]
- Open-sourced the full pipeline, gathering **120+ GitHub stars** and broad community interest.

Music and AI Lab, Academia Sinica Jul 2019 – Sep 2023
Research Assistant (Advisor: Prof. Yi-Hsuan Yang) Taipei, Taiwan

- Pioneered **loop-based music AI research** spanning compatibility assessment, generation, and benchmarking. [C3] [C4] [C5]
- Mentored **5+ junior researchers**, co-authoring 4 papers and establishing reproducible research codebases.

OPEN-SOURCE SOFTWARE & TOOLKITS

OpenMIRLab

Nov 2025 – Present

Founder & Lead Maintainer

Open-Source

- **Packaged reproducible inference suites for 30+ SOTA MIR models** (source separation, transcription, chord estimation, beat tracking), resolving upstream dependency decay across PyTorch and TensorFlow.
- **Standardized unified inference APIs and checkpoint registries**, stripping heavy training dependencies into lightweight, pip-installable packages with verified model outputs.

WORK EXPERIENCE

Rhythm Culture Corporation

Jul 2023 – Present

Technical Lead

Taipei, Taiwan

- **Architecting Rytho, a local-first creative workspace ("notes with ears") empowering Human-AI co-creation**, serving as the central hub coordinating autonomous agent abilities within the **Rytho ecosystem**.
- **Led R&D on LLM-driven music agents**, developing conversational feedback agents for music production and reasoning systems for chord analysis. [\[W1\]](#) [\[W2\]](#)
- **Managed engineering teams for government R&D projects** (TAICCA, Taipei City) and **TMC-CL1**, a web music interface commissioned by Taipei Music Center.

Zone Sound Creative Corporation

Mar 2019 – Aug 2019

Interactive Engineer

Taipei, Taiwan

- **Architected interactive web audio interfaces and networked performance systems**, developing a vocal sampler for Vocal Asia (Japan) [\[A2\]](#) and a p5.js/Node.js WebSocket engine bridging mobile audiences to Max/MSP stage synthesis [\[A4\]](#).
- **Engineered generative audio-visual systems** in collaboration with electronic music artists, delivering responsive sound-visual synchronization pipelines for MIDEM (France). [\[A3\]](#)

PUBLICATIONS

C1 FlueBricks: A Construction Kit of Flute-like Instrument for Acoustic Reasoning

Bo-Yu Chen, Chiao-Wei Huang, Lung-Pan Cheng

ACM Conference on Human Factors in Computing Systems (CHI 2026)

Acceptance ~25%

[TAICHI 2025 · Best Paper](#)



W1 Enhancing Automatic Chord Recognition through LLM Chain-of-Thought Reasoning

Chih-Cheng Chang, Bo-Yu Chen, Lu-Rong Chen, Li Su

ISMIR 2025 Workshop on Large Language Models for Music (LLM4Music)



W2 AI TrackMate: Finally, Someone Who Will Give Your Music More Than Just "Sounds Great!"

Yi-Lin Jiang, Chia-Ho Hsiung, Yen-Tung Yeh, Lu-Rong Chen, Bo-Yu Chen

Conference on Neural Information Processing Systems (NeurIPS 2024) – Creative AI Track

Acceptance 31%



C2 Automatic DJ Transitions with Differentiable Audio Effects and Generative Adversarial Networks

Bo-Yu Chen, Wei-Han Hsu, Wei-Hsiang Liao, Marco A. Martínez Ramírez, Yuki Mitsufuji, Yi-Hsuan Yang

IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2022)

Acceptance 47%



C3 Exploiting Pre-trained Feature Networks for Generative Adversarial Networks in Audio-Domain Loop Generation

Yen-Tung Yeh, **Bo-Yu Chen**, Yi-Hsuan Yang

International Society for Music Information Retrieval Conference (ISMIR 2022)

Acceptance 43%



C4 A Benchmarking Initiative for Audio-Domain Music Generation Using the Freesound Loop Dataset

Tun-Min Hung, **Bo-Yu Chen**, Yen-Tung Yeh, Yi-Hsuan Yang

International Society for Music Information Retrieval Conference (ISMIR 2021)

Acceptance 34%



C5 Neural Loop Combiner: Neural Network Models for Assessing the Compatibility of Loops

Bo-Yu Chen, Jordan Smith, Yi-Hsuan Yang

International Society for Music Information Retrieval Conference (ISMIR 2020)

Acceptance 38%



INTERACTIVE ART & INSTALLATIONS

A1 Iridescent Vision 2019

Developed a web-based interactive music video and light installation exploring human civilization and imagined future technology, with DJ Sonia Calico and visual artist Celso Urroz · FINAL Club (Taipei)

A2 Taptap 2019

Developed a web-based musical interface that lets visitors play music with a cappella samples, in collaboration with O-KAI SINGERS · Vocal Asia Festival (Kagawa, Japan)

A3 FusionTaiwanBeat 2019

Built a web-based interactive audiovisual system showcasing Taiwanese electronic music, with DJ Sonia Calico and visual artist Celso Urroz · MIDEM (Cannes, France)

A4 Duplex Transmission 2019

Built a web-based audience-performer interaction system using p5.js, Tone.js, Node.js (WebSockets), and Max/MSP, with new media artist Po-Hao Chi · Songshan Cultural Park (Taipei)

A5 Mimo No.Tent 2019

Created a virtual planetarium that brings a starry sky inside a tent, with designers Yi-Yun Liao and Jing-Ting Huang · TaiwanTech Design Grad Show (Taipei, Taiwan)

A6 YuYu 2018

Developed an AR service for delivering customized messages through Chinese pastries, with designer Ya-Cin Siao · TaiwanTech Design Grad Show (Taipei, Taiwan)

A7 Message Container 2017

Designed a breath-driven immersive installation providing haptic, sound, and visual feedback, with new media artist Wen-Fei Zhang · TaiwanTech Design Grad Show & Venue (Taipei, Taiwan)

ACADEMIC SERVICE & REVIEWING

International Society for Music Information Retrieval Conference (ISMIR)	2020 – 2026
IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)	2025
Conference on Neural Information Processing Systems (NeurIPS) – Creative AI Track	2025

TEACHING EXPERIENCE

BLND Creative Coding School	Aug 2020 – Jul 2023
Instructor	Taipei, Taiwan
Taught generative AI art (GANs, Stable Diffusion workflows) and physical computing with Arduino to creative practitioners.	
OrangeApple Children Programming School	Jul 2015 – Sep 2019
Instructor	Taipei, Taiwan
Taught algorithms and C++/JS to K-12 students; coached high school students for the APCS examination.	

EXTRACURRICULAR ACTIVITIES

Musical Theater: Rose of the Endless Night  	Feb 2015 – Jun 2015
Director, Music Theater Course Final Public Performance (NTUST)	Taipei, Taiwan
Authored a stage adaptation of a Taiwanese musical and directed 60+ cast and crew members for a live stage production with 400+ audience.	
NTUST Men's Football Team	Jul 2016 – Jul 2017
Vice Captain, Position: Defensive Midfielder / Right-Back	Taipei, Taiwan
- Coordinated squad training and match operations across 30 collegiate athletes and coaching staff. - Led team to 4th Place in the National Inter-University Football Tournament and 5th Place in the National Futsal Tournament.	